

Ontario Winters and El Niño

A four-station analysis of how past El Niño winters looked in Ontario

Prepared	September 19, 2026
Repository name	ontario-elnino-winters
Level and duration	Beginner to lower-intermediate; 4 weeks (September 21 to October 18, 2026); about 10 to 12 hours per week
Season and region	Winter (December to February), Ontario, Canada
Budget	\$0 (all data and tools are free)

1. The Story and Purpose

One-line pitch: A very strong El Niño is developing for winter 2026-27. Before it arrives, this project looks back at about 45 winters in four Ontario cities to see what El Niño winters actually looked like here.

NOAA issued an El Niño Advisory on August 13, 2026, with a greater than 90% chance of a very strong event through winter 2026-27. Headlines say "milder winter for Canada," but those statements are broad. Windsor and Thunder Bay have very different winters, and Ontario's response to El Niño is less clear than in western Canada. This project asks a simpler, local question: **what did past El Niño winters look like in Ontario, from south to north?**

Purpose and users

- **Municipal winter maintenance teams:** to see whether past El Niño winters had fewer snow days or less very cold weather.
- **Ski and winter tourism operators:** to see how snowfall differed, especially in the north.
- **The public:** a plain answer to "will El Niño make my winter milder?" for their own city.
- **You, as the analyst:** a finished project that takes a real question from raw data to a clear, honest report.

What kind of analysis is this?

It is mainly **comparative (composite) analysis**: comparing groups of winters by ENSO class. It also uses descriptive statistics, a time-series trend line, bootstrap confidence intervals, and a south-to-north comparison. It studies climate *variability* (natural year-to-year swings), not long-term climate *change*, and it shows associations, not proof of cause. It is historical and does not forecast winter 2026-27.

2. Research Questions

1. Were El Niño winters warmer than neutral winters at each station?
2. Did total snowfall differ between El Niño winters and other winters?
3. Did the number of very cold days (minimum temperature at or below -20°C) change?
4. Does the effect differ between the south (Windsor) and the north (Sudbury, Thunder Bay)?

3. Stations

Region	Proposed station	Why it was chosen
Southwest	Windsor A	Mildest winters, the warm end of the range
East	Ottawa Macdonald-Cartier Int'l A	Cold, snowy eastern Ontario with a long airport record
Northeast	Sudbury A	Central-north station with heavy snow
Northwest	Thunder Bay A	Coldest and snowiest; strongest contrast with Windsor

Selection criteria

- As many of the 45 winters as possible with at least 85 of 90 valid days.
- A clean record: one continuous station, or a simple two-station switch.
- Snowfall reported at that station, not only temperature.
- Spread across Ontario, since the south-to-north pattern is a key question.

Why Toronto is not in the main four

- **Split record.** Its data sits under two stations: Toronto Lester B. Pearson Int'l A (the older record) and Toronto Intl A, which appears to start around mid-2013. The two must be stitched together, and a small shift in temperatures at the switch date is possible. That adds work and uncertainty to a 4-week beginner plan.
- **Geographic overlap.** Toronto sits between Windsor and Ottawa in southern Ontario. The four proposed stations cover the southwest, east, northeast, and northwest, which tests the south-to-north question better.
- **Not a rejection.** Toronto is still a good station, and adding it later takes one new row in stations.csv. If its stitched record turns out to be more complete than a proposed station, swap it in. The other proposed stations may also have split records, which the completeness check will show.

Backups and options

- **London A:** southwestern snowbelt. **Sault Ste. Marie or North Bay:** northern alternatives to Sudbury.
- **Toronto (optional fifth station):** older record Toronto Lester B. Pearson Int'l A (Climate ID 6158733, StationID 5097); newer record Toronto Intl A (Climate ID 6158731, StationID 51459).
- Long records are often split across station IDs, so any candidate may need checking. Station ID numbers are internal and can change; the 7-digit Climate ID is the official identifier.

Not yet verified: record completeness for each station. Shortlist 6 to 8 candidates, check each one's period of record for daily data, download them, and keep the four most complete stations that still cover different parts of Ontario.

4. Data Sources and How to Collect Them

Dataset	Provider	Use	Where
Daily climate data (max, min, mean temperature; total snow)	Environment and Climate Change Canada	Winter measures for each station	Climate Data Extraction Tool (Historical climate and river data, then Daily climate data), or the bulk download URL below
RONI (official ENSO index)	NOAA Climate Prediction Center	Labelling each winter El Niño, neutral, or La Niña	cpc.ncep.noaa.gov/data/indices/RONI.ascii.txt
ONI (optional check)	NOAA Climate Prediction Center	Compare classification with the older index	cpc.ncep.noaa.gov/data/indices/oni.ascii.txt
AHCCD (optional)	Environment and Climate Change Canada	Cross-check temperature trends	Value-added historical climate data products. Snowfall is water content in mm, not cm.

Bulk download URL (one request returns one station-year of daily data)

```
https://climate.weather.gc.ca/climate_data/bulk_data_e.html
?format=csv&stationID=<ID>&Year=<YYYY>&Month=1&Day=1
&timeframe=2&submit=Download+Data
```

Collection steps

1. Create data/raw/ and config/stations.csv with name, region, climate_id, and station_id.
2. Find each station's StationID (in the page URL) and Climate ID, and note its years of daily data.
3. Check for split records and add a second row for any earlier or later station ID.
4. Download one file by hand and confirm it has about 365 daily rows and a snowfall column.
5. Download the full period, December 1981 to February 2026, using the extraction tool or a short Python loop.
6. Download the RONI text file, keep the DJF rows, and save a raw copy with the download date.
7. Verify: count files per station, check rows per year, and read one file into pandas (try encoding utf-8-sig, or latin-1).
8. Record each source, URL, station ID, Climate ID, and download date in the README.

5. Study Design

Period: winters 1981-82 to 2025-26 (about 45 winters). **Winter:** December, January, and February, named after the year in which February falls. December 1997 to February 1998 is winter "1997-98" and is stored as `winter_year = 1998`. In the NOAA text files, the DJF row for year Y matches `winter_year = Y` directly.

Classifying winters

Class	DJF index value
El Niño	+0.5 or higher
Strong El Niño (a subset of El Niño)	+1.5 or higher
Neutral	between -0.5 and +0.5
La Niña	-0.5 or lower

Use RONI (NOAA's official index since February 2026) and state that choice in the report. The choice matters, because the strong-winter list depends on the index:

Group	Strong El Niño winters (DJF at or above +1.5)
Strong in both ONI and RONI	1982-83, 1991-92, 1997-98, 2009-10, 2015-16
ONI only	2023-24 (about 1.9 in ONI, but only about 1.2 in RONI)
RONI only	1986-87 (about 1.6 in RONI, but about 1.2 in ONI)

These values were read from NOAA tables based on the older ERSSTv5 data, and NOAA switched to ERSSTv6 in August 2026. Recheck against the current text files. 2009-10 sits right on the +1.5 cutoff, so use the two-decimal values from the text files. Sanity check: RONI for DJF 2026 is about -0.88, so winter 2025-26 should classify as La Niña.

Measures per station, per winter

- **Mean winter temperature** and its **anomaly** (difference from that station's 1991-2020 winter average)
- **Total snowfall** (cm) and **snow days** (days with at least 1 cm)
- **Very cold days** (minimum temperature at or below -20°C)

Data quality rule: keep a station-winter only if at least 85 of 90 days have valid temperature readings. Fill temperature gaps of up to 3 consecutive days by linear interpolation and flag them. Do not fill missing snowfall. Log every dropped station-winter and the reason.

6. Analysis Plan

Compare winter types. For each station, average every measure in El Niño, neutral, and La Niña winters and calculate the El Niño minus neutral difference. Show strong El Niño winters separately, remembering there are only about five.

Five charts

1. Box plot of temperature anomaly by ENSO class, one panel per station
2. Bar chart of the El Niño minus neutral temperature difference by station, with confidence intervals
3. Timeline of winter temperature anomaly, 1982 to 2026, El Niño winters highlighted, with a trend line
4. Box plot of total snowfall by ENSO class
5. Bar chart of very cold days by ENSO class

Uncertainty. Calculate a bootstrap 95% confidence interval for each difference by resampling winters with replacement 10,000 times. If an interval includes zero, describe the result as inconclusive. **Warming trend.** Use anomalies rather than raw temperatures and show the trend line, so long-term warming is not mistaken for an El Niño effect.

7. Key Performance Indicators

Analysis KPIs (dashboard and report)	Project KPIs (progress self-check)
Temperature anomaly difference, El Niño minus neutral (°C), per station	Stations completed: target 4, minimum 3
Snowfall difference (cm or %) per station	Data completeness: % of station-winters passing the 85 of 90 day rule
Snow days difference per winter	SQL row counts and averages match the pandas output
Very cold days difference per winter	Charts finished: target 5, minimum 3
Whether each confidence interval includes zero	Every reported difference has a confidence interval (100%)
Number of winters in each group	Weekly checkpoints hit: 4 of 4

8. Dashboard and Using the Results

- **Map view:** the four stations with an ENSO class filter and a card showing average temperature anomaly.
- **Comparison view:** El Niño minus neutral difference by station and measure.
- **Timeline view:** winter anomalies by year, coloured by ENSO class, with a station selector.

Recommendations should be framed as things to consider, with the uncertainty shown. For example, if El Niño winters show fewer snow days, maintenance and winter tourism may consider a lower snow and salt budget with a reserve for a big storm. If differences are small, plan as usual, and keep cold-snap preparations in place. Natural gas heating is not captured in this data, so avoid claims about total energy use.

9. Tech Stack and Repository

Layer	Tool	Purpose
Language	Python 3.11 or newer	Loading, cleaning, analysis
Data handling	pandas, NumPy	Clean and summarize tables, resampling
Download	requests	Automate station file downloads
Charts	matplotlib, seaborn	Box plots, timelines, comparisons
Database	SQLite	Store cleaned data and run SQL summaries
Notebook and editor	Jupyter or Google Colab; VS Code	Step-by-step code and scripts
Dashboard	Tableau Public or Power BI (Windows)	Interactive final visuals (export tables to CSV first)
Version control	Git and GitHub	Save versions and publish the project

```
pip install pandas numpy matplotlib seaborn requests jupyter
```

Database tables

Table	Columns
stations	station_id, name, region, climate_id, latitude, longitude
daily_obs	station_id, date, tmax, tmin, tmean, snowfall_cm
enso_monthly	year, season, roni
winter_summary	station_id, winter_year, mean_temp, temp_anomaly, snowfall_total_cm, snow_days, days_below_m20, enso_class

```
ontario-elnino-winters/  
  config/stations.csv  
  data/raw/    data/processed/  
  sql/winter_summary.sql  
  notebooks/  src/ (load.py, clean.py, summarize.py)  
  dashboard/  reports/  README.md  requirements.txt
```

10. Four-Week Roadmap

Each week has six working days of about 2 hours, plus Day 7 as a buffer. Do not skip the buffer days.

Week	Focus	Key tasks	Checkpoint
1 Sep 21-27	Setup, first station, first results	Install Python and Jupyter, create the GitHub repo and folders. Download one station and the RONI file. Clean the data, build winter measures and anomalies, label winters by ENSO class, make a box plot and timeline.	Mini-project complete for one station
2 Sep 28-Oct 4	Four stations and SQL	Turn code into three functions and create stations.csv. Check the shortlisted stations, confirm the final four (decision point). Write the gap log. Load SQLite, reproduce the winter summary in SQL, and compare with pandas.	Combined table; SQL matches pandas
3 Oct 5-11	Charts, uncertainty, findings	Produce the five charts. Build the differences table, including a strong El Niño subset. Add bootstrap intervals. Write 5 to 8 findings, each with a caveat, in findings.md.	Five charts, intervals, findings list
4 Oct 12-18	Dashboard, report, polish	Export CSVs and build three dashboard views. Write the 4 to 6 page report. Finish the README and requirements.txt. Check every item in the definition of done.	Definition of done met

If you fall behind, cut in this order

1. The dashboard (replace it with the matplotlib charts).
2. One station (go to three, decided by the end of Week 2).
3. SQL depth (keep only the single reproduced summary query).

Never cut the winter summary, the ENSO comparison charts, the bootstrap intervals, or the limitations section. This is a tight schedule for a beginner. If you can only manage 6 to 8 hours a week, use the same plan over 6 weeks.

11. Risks and Limitations

Risk	Effect	What to do
Only about five strong El Niño winters	Uncertain results	Show ranges, avoid strong claims, use bootstrap intervals
Strong-winter list depends on the index (RONI or ONI)	Different groups, different results	State the index used; optionally repeat with ONI
Missing data or split station records	Biased averages	85 of 90 day rule, gap log, station backups, record the switch date
Warming trend	May look like an El Niño effect	Use anomalies and a trend line
Weak El Niño signal in Ontario	Small or no difference	Report it plainly; a null result is a valid finding
Many stations and measures	Some differences appear by chance	Focus on patterns consistent across stations
Reading association as cause	Overclaiming	State clearly that the analysis is not causal
Tight 4-week schedule	Rushing, errors	Use buffer days and the cut list

Value and limits: Environment and Climate Change Canada and NOAA already study El Niño's effects on Canada with far more data and better models. This project is a transparent, reproducible station-level analysis that provides historical context, not a forecasting tool. A clear "the effect is small and uncertain" is a valid and useful result.

12. Definition of Done

- At least three stations (ideally four) are cleaned and summarized.
- The winter summary table exists both in pandas and in SQL.
- At least three of the five charts are finished.
- Every reported difference includes a bootstrap confidence interval.
- The report states what was found, how certain it is, and what the data cannot show.
- The GitHub repository has a README that lets someone else rerun your work.

Optional extensions

- Permutation test as a second significance check; repeat the comparison after removing the linear warming trend.
- Add Toronto, London, or a BC or Alberta station (where the El Niño signal is expected to be stronger).
- Add electricity demand from IESO, Great Lakes ice cover, or the PDO index.
- Compare what actually happens in winter 2026-27 with the historical El Niño pattern once the season ends.